

ADFORGE AI — PROJECT DOCUMENTATION

Technology Stack Definition

- **Date:** 7th July 2026
- **Development Profile:** Sunehri Sharma
- **Maximum Marks:** 2 Marks

Proposed Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Modern Dark-Themed Glassmorphism UI), and vanilla JavaScript (Fetch API)	Chosen to build a responsive, single-page dashboard with fast rendering. Vanilla JavaScript dynamically handles the asynchronous API states, updates loading progress animations, and enables quick copy-to-clipboard buttons.
2	Backend / Server-Side	Python / Flask	A lightweight, fast micro-framework ideal for local development. Flask excels at handling simple routing (app.py), parsing form payloads into structured JSON formats, and securely executing third-party API configurations in Python.
3	Database / Data Storage	Zero-Database Architecture / Local System Environment Validation (.env)	Since the system generates ad variants dynamically and does not require persistent user profiles, we bypass traditional database storage. This ensures absolute user data privacy by not logging custom business briefs to disk.
4	Cloud / Hosting / Deployment	Localhost Development Port (127.0.0.1:5000) with Secure Mirror	Allows rapid local coding in VS Code. Integrating the pyngrok wrapper exposes the local Flask port to a

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		Tunneling via pyngrok	secure, public HTTPS URL so clients or evaluators can test the generation engine remotely without server setups.
5	Version Control & CI/CD	Git & GitHub	Used for complete code tracking, file change management, and establishing safe repository branch systems (main). It also hosts all project assets, code files, and structured milestone folders.
6	Third-Party APIs / Other Tools	Groq Cloud API SDK (LLaMA 3.3-70B-Versatile model)	The core computational engine of the project. This high-performance model processes structured marketing briefs to return conversion-focused multi-platform copies, slogans, and A/B suggestions under strict JSON constraints in seconds